Tuning report: tropical / FloydWarshall

2026-10-22T19:17:30

1 Decision

Approving freezes `fishr.lambda` = 1e+04 into every cell of the grid (frozen hash `b4e69c01851f`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 10000.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: the winner is not distinguishable from fishr.lambda=10: the paired gap's 95% CI is [-0.000669, 0.00994]
- **review**: fishr.lambda=10000 is the largest value searched: the optimum may lie beyond it, so widen the space
- **review**: fishr.lambda=10: fishr.lambda * fishr_penalty_final is 0.0011 of id_val_loss (median over runs); the term barely enters the loss
- **review**: fishr.lambda=100: fishr.lambda * fishr_penalty_final is 0.00076 of id_val_loss (median over runs); the term barely enters the loss
- **review**: fishr.lambda=1000: fishr.lambda * fishr_penalty_final is 0.0042 of id_val_loss (median over runs); the term barely enters the loss
- **review**: 12 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 12 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`

  - `model_type=adaptive head=seq2seq objective=fishr`
  - `model_type=adaptive head=pointer objective=fishr`
  - `model_type=tropical head=seq2seq objective=fishr`
  - `model_type=tropical head=pointer objective=fishr`
- Search-only overrides (never frozen): `{'optim': {'epochs': 20}}`. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	`fishr.lambda`	runs	div.	mean	95% CI	SD	range	
2	10	12/12	0	0.07947	[0.0751, 0.0844]	0.00847	[0.0733, 0.0963]	
3	100	12/12	0	0.07981	[0.0758, 0.0849]	0.00907	[0.0732, 0.0963]	
4	1000	12/12	0	0.07986	[0.0754, 0.0849]	0.00885	[0.073, 0.0965]	
1	1e+04	12/12	0	0.07517	[0.0736, 0.0772]	0.00344	[0.0716, 0.0848]	**selected**

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=10`	12	0.004303	$[-6.69 \times 10^{-4}, 0.00994]$	7	5	0	0.7744
`fishr.lambda=100`	12	0.004646	$[-7.17 \times 10^{-4}, 0.0107]$	6	6	0	1.0000
`fishr.lambda=1000`	12	0.004691	$[-5.71 \times 10^{-4}, 0.0106]$	5	7	0	0.7744

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=1000
`model_type=adaptive head=seq2seq`	fishr.lambda=10000
`model_type=tropical head=pointer`	fishr.lambda=10000
`model_type=tropical head=seq2seq`	fishr.lambda=1000
`model_type=vanilla head=pointer`	fishr.lambda=1000
`model_type=vanilla head=seq2seq`	fishr.lambda=10000

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

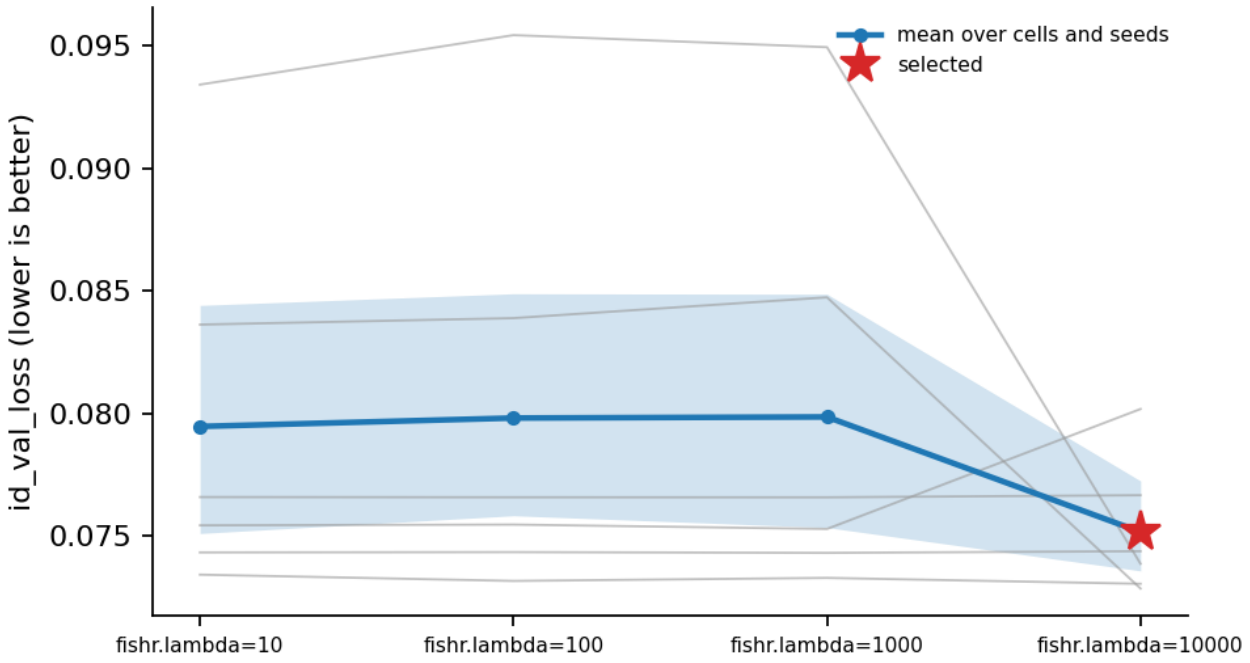


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.95	0.606	1.54×10^{-5}	19
fishr.lambda=100	0.95	0.606	1.18×10^{-6}	19
fishr.lambda=1000	0.95	0.606	3.74×10^{-7}	19
fishr.lambda=10000	0.952	0.613	1.74×10^{-7}	19

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

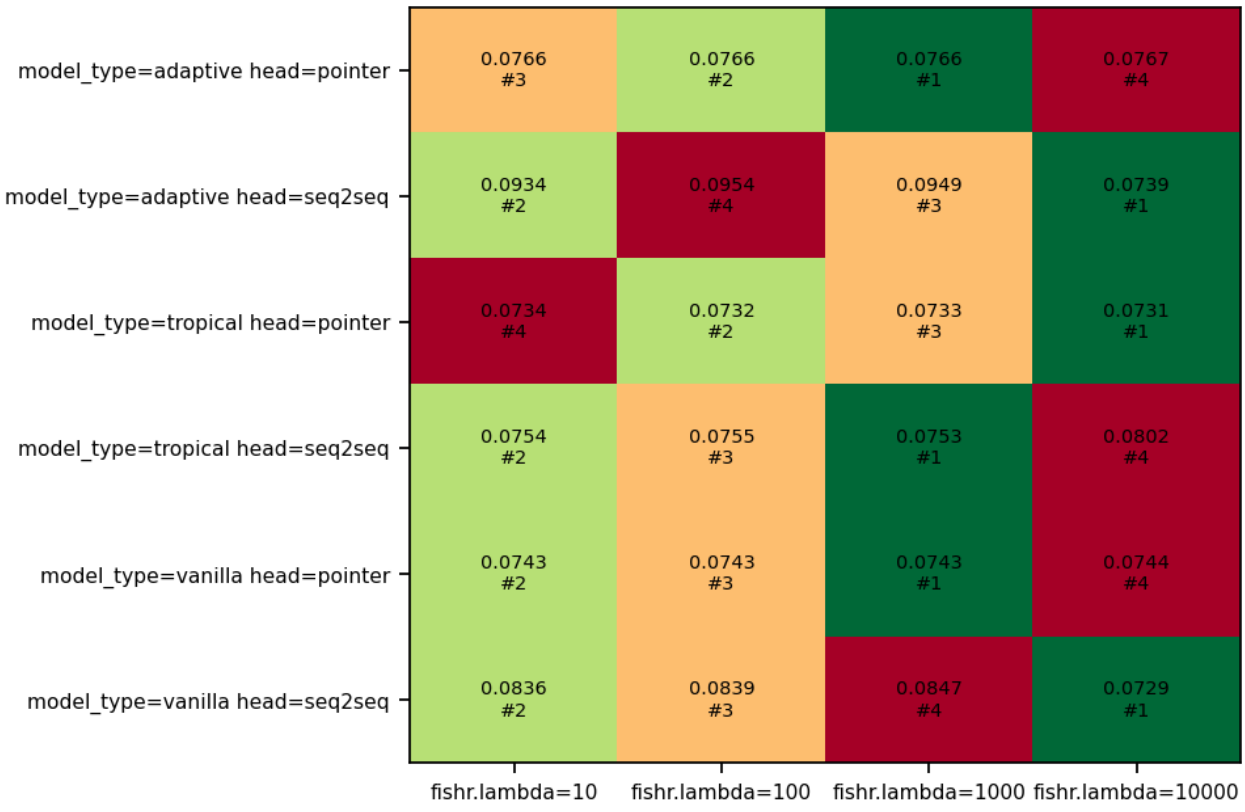


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

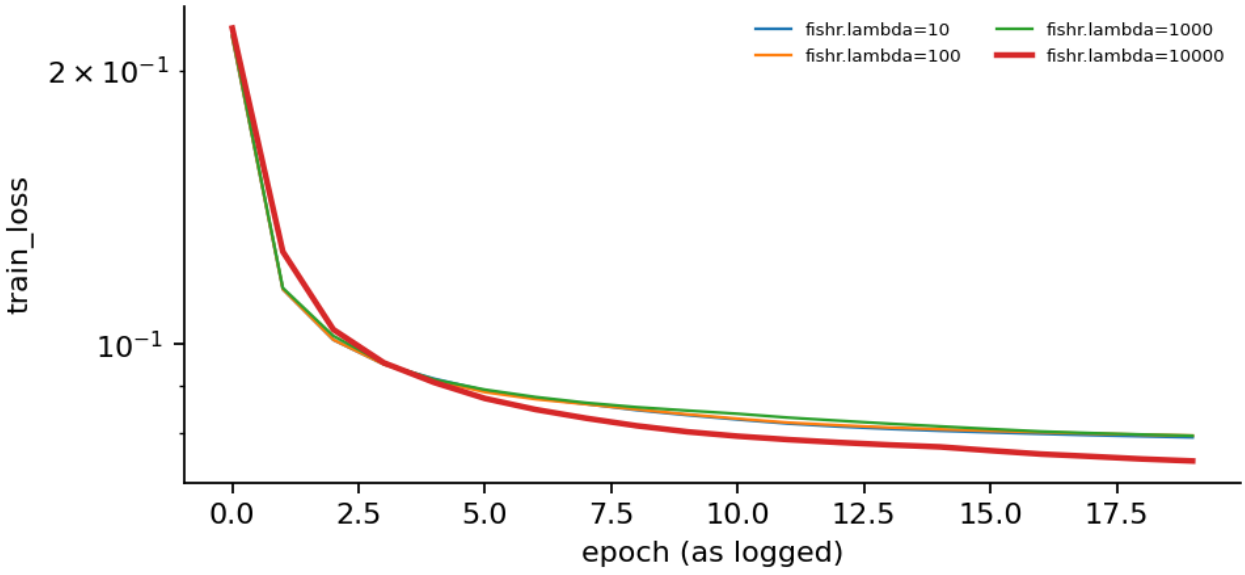


Figure 3: `train_loss` per candidate.

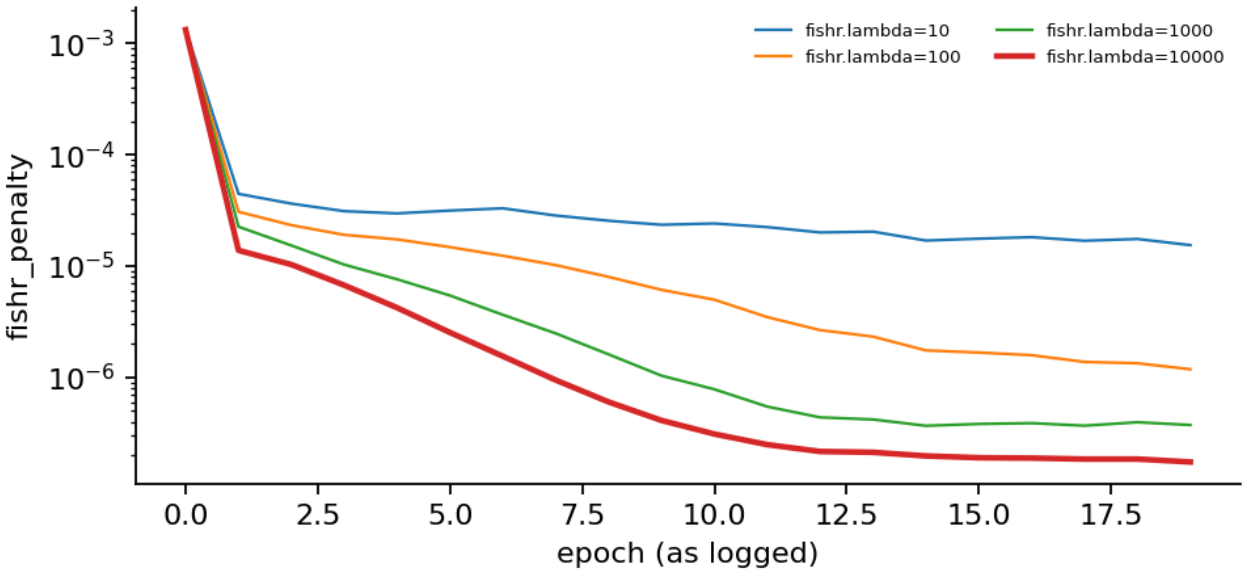


Figure 4: `fishr_penalty` per candidate.

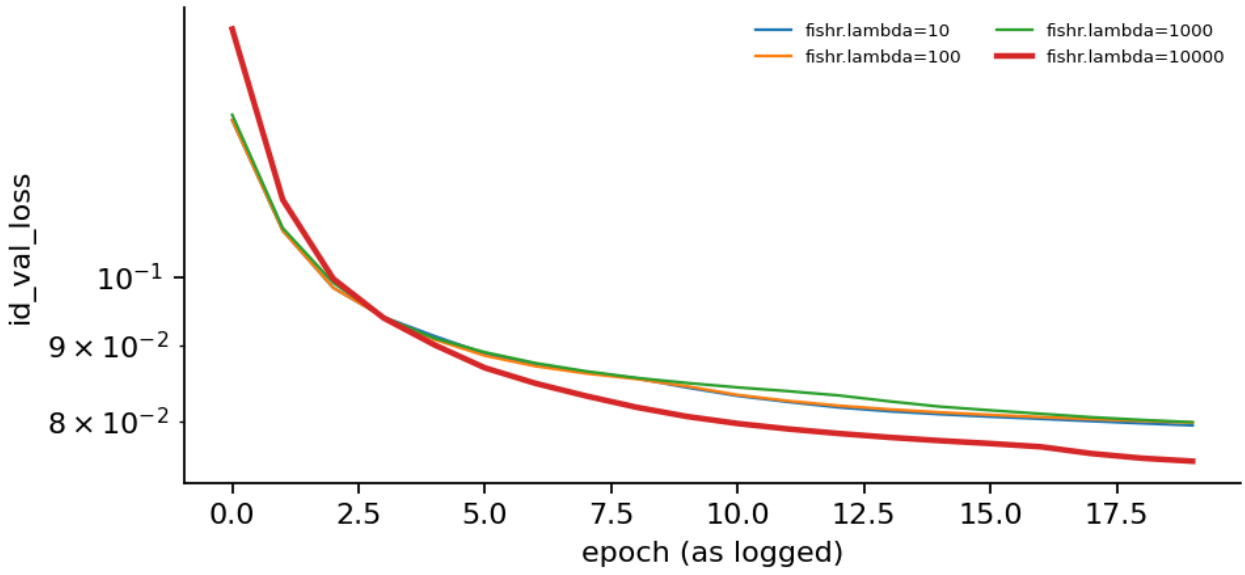


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	b5b535e95b3b	12	3.42e+04	e5ccce8034	2.8.0+cu128
fishr.lambda=100	01066968f82c	12	2.68e+04	e5ccce8034	2.8.0+cu128
fishr.lambda=1000	4668f1802c0f	12	1.96e+04	e5ccce8034	2.8.0+cu128
fishr.lambda=10000	15db75e05873	12	3.61e+04	e5ccce8034	2.8.0+cu128

Store: `campaigns/tropical/FloydWarshall/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile FloydWarshall`
Approve: `expkit approve --config configs/tropical.yaml --profile FloydWarshall`

The base the grid would run with.

```
task:
  name: FloydWarshallDataset
  value_range:
  - 1
  - 15
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 20
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  ood_value_range:
  - 16
  - 30
  ood_noise_range:
  - 1
  - 10
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 8
  eval_batch_size: 8
fishr:
  lambda: 10000.0
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16
fishr.lambda=10	0.901	0.0511	0.588	0.44	0.605	0.00439	0.916	0.09	0.893	0.0123
fishr.lambda=100	0.901	0.0517	0.479	0.397	0.601	0.00439	0.915	0.0912	0.892	0.0123
fishr.lambda=1000	0.901	0.0514	0.544	0.489	0.607	0.00468	0.915	0.0904	0.892	0.0123
fishr.lambda=10000	0.908	0.0531	0.509	0.427	0.589	0.00433	0.92	0.093	0.901	0.0131